

Brainstorming & Idea Prioritization

Phase 1 — Brainstorming & Ideation

Project: **PocketSmart AI — Your Smart Budget & Recommendation Assistant**

1. Team

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2. Where the Idea Came From

Almost everyone eventually faces a moment where a fixed amount of money has to be split across many things: furnishing a new home, organising a birthday or wedding function, or buying jewelry for a festival. The usual approach is to open many browser tabs, compare prices by hand, ask friends and family, and keep a rough total in a notebook or phone. It is slow, easy to get wrong, and often ends with overspending on one item and having too little for another.

We started the brainstorming session with a simple question: **"What if a person could just type their total budget and requirements, and get a ready-made plan showing what to buy, roughly how much each thing costs, and where to buy it?"** Recent generative AI models such as Google Gemini are good at exactly this kind of reasoning, so the idea became an AI-driven budget assistant.

3. Problem Statement

People planning purchases or events in India face four connected problems:

- **Splitting a budget** — they do not know how much of the total should go to each category (for example, how much of a home budget goes to lighting versus furniture).
- **Knowing realistic prices** — price information is scattered across many shops and websites, so it is hard to judge whether a plan is realistic.
- **Finding where to buy** — after deciding what to buy, they still have to search each store separately.
- **Keeping track** — plans are made on paper or in chats and are lost or forgotten.

4. Proposed Solution

PocketSmart AI is a web application in which the user:

Registers and logs in securely.

Chooses a planner: Home, Party or Jewelry.

Enters a total budget plus a few simple requirements (for example, number of fans, number of guests, or the occasion).

Receives an AI-generated plan: a category-wise budget breakdown, suggested items with estimated prices and quantities, the remaining budget, and extra suggestions.

Clicks direct shopping or service links for every suggested item.

Comes back later to see previous recommendations on the dashboard and history page.

5. Brainstorming Session — Ideas Considered

We listed many possible project ideas and evaluated each against three questions: Is it useful to ordinary people? Can we build it in the available time? Does it make good use of AI?

Ideas that were considered and shortlisted

- **Home budget planner** — the user enters a budget and how many lights, fans, furniture pieces and dining tables they need, and which rooms (living room, kitchen, bedroom) are involved. Very practical, and product categories are easy to search online.
- **Party budget planner** — the user enters the party type, number of guests, venue type and whether catering, decoration and entertainment are needed. Covers both products (decor) and services (food delivery, event tickets).
- **Jewelry budget planner** — the user enters an occasion and preferences, and can optionally upload a photo of their outfit so the AI can match jewelry to colours and style. This is the most creative use of AI because it uses image understanding.

Ideas that were discussed but postponed

- Automatic live price tracking across shops — needs shop APIs or scraping, which is outside our scope.
- Payment and ordering inside the app — adds legal and security complexity.
- A mobile application — a responsive web app is faster to build and works on phones anyway.
- Group budgeting where several family members edit one plan — needs a real database and permissions.

Final decision: build one application with three planners (Home, Party, Jewelry), with login, a dashboard and a history page. Everything else was moved to the future roadmap.

6. Target Users

- **Families setting up or renovating a home** who want a sensible split between lighting, fans, furniture and dining needs.
- **Students and young professionals** organising birthday parties or small events on a tight budget.
- **Anyone shopping for jewelry** for a wedding, festival or gift who wants suggestions that fit both their budget and their outfit.
- **First-time buyers** who do not know typical prices and need a starting point.

7. Example Use Cases

These are illustrative scenarios used during brainstorming to shape the design:

- A family moving into a new flat has ■1,00,000 for lights, fans and furniture. They want to know how to divide it and what to search for.
- A college student is planning a birthday party for 30 friends with a ■20,000 budget and needs a split across food, decoration and entertainment.
- A user is attending a wedding, uploads a photo of their saree, sets a ■30,000 budget, and wants jewelry that suits the colours.

8. What Makes PocketSmart AI Different

- **Budget-aware output** — the AI is instructed to keep total cost within the entered budget and to report what remains.
- **Structured results** — the AI is asked to reply only in a fixed JSON format, so the application can process the answer, not just display text.
- **Image-aware jewelry advice** — an uploaded outfit photo is analysed for colours, style and formality.
- **Action, not just advice** — every item comes with search links to Amazon India, Flipkart, IKEA India, Myntra, Swiggy, Zomato, BookMyShow, Tanishq or CaratLane, depending on the planner.
- **Personal history** — each user's past recommendations are kept and shown again on the dashboard and history pages.
- **India-focused** — prompts and platforms are chosen for Indian users and INR budgets.

9. Initial Strengths, Weaknesses, Opportunities and Threats

Strengths

- Solves a real, common problem.
- Simple, modular design that a small team can build and explain.
- Uses modern tools (FastAPI, Gemini, JWT).

Weaknesses

- Prices are AI estimates, not live shop prices.
- Data is stored in memory in this version, so it is lost on restart.

Opportunities

- Add more planners (wedding, travel, gadgets).
- Add a database, price comparison and multi-language support.

Threats

- Changes or retirement of the AI model or SDK.
- API usage limits or cost.
- Shopping sites may change their search URL formats.

10. Success Criteria Defined at This Stage

- A new user can register, log in and reach the dashboard.
- Each of the three planners returns a readable plan for a reasonable input.
- Every suggested item has working shopping links.
- Passwords are never stored in plain text and protected pages need login.
- The team can explain the architecture and demonstrate all features.

11. Expected Outcome

A working web application named PocketSmart AI with registration and login, a personal dashboard, three AI-powered planners, shopping links, and a history page, plus complete documentation covering all eight project phases.